Camera and lens: complete derivation

Project RescueSwarm · NIDAR 2026–27 Track 1 · generated from `tools/sizing-model/camera_optics.py`

1. The part, and a discrepancy

The bill of materials buys an **Arducam IMX477** with a 6 mm S-mount lens. Sony specifies the IMX477 as type 1/2.3: 4056 × 3040 at a **1.55 µm** pixel pitch, 7.9 mm diagonal.

The sizing chapter was not written against that sensor. `sizing-calculations.md` §8 states the baseline as "1/1.8 in, 4056 × 3040, 7.4 × 5.6 mm, **1.82 µm** pitch". The pixel *count* matches and the pixel *size* does not. Every optical quantity below scales with pitch, so none of the published figures describe the camera being bought. Both cases are computed here.

2. Sensor geometry

Active area follows from count and pitch:

$$w = N_x\, p, \qquad h = N_y\, p, \qquad d = \sqrt{w^2 + h^2}$$

$$w = 4056 \times 1.55\,\mu\text{m} = 6.287\,\text{mm}, \quad h = 3040 \times 1.55\,\mu\text{m} = 4.712\,\text{mm}, \quad d = 7.86\,\text{mm}$$

3. Field of view

The lens forms the image at its focal length, so each half-field is the arctangent of half the sensor over f — exact, with no small-angle approximation:

$$\theta = 2\arctan\left(\frac{s}{2f}\right)$$

$$\text{HFOV} = 2\arctan\frac{6.287}{12.0} = 55.3^\circ, \qquad \text{VFOV} = 42.9^\circ, \qquad \text{DFOV} = 66.4^\circ$$

One pixel subtends 14.80 mdeg, which is the quantity that sets the blur tolerance in §6.

4. Ground sample distance and footprint

Sensor and ground form similar triangles about the lens, so

$$\text{GSD} = \frac{p\,H}{f}, \qquad W = 2H\tan\frac{\text{HFOV}}{2}, \qquad S = W\,(1-\ell)$$

for altitude H, pitch p, focal length f and sidelap $\ell = 30\,\%$. Sidelap exists because attitude wobble, altitude error and lens distortion all move the real footprint around; at zero sidelap a gap between passes is an undetected survivor.

AGL (m)	**GSD** (cm/px)	**Survivor** (px)	**Footprint** (m)	**Spacing** (m)	**Lines**	**Sweep** (s)
30	0.78	219	31.4 × 23.6	22.0	9	253
40	1.03	165	41.9 × 31.4	29.3	7	196
60	1.55	110	62.9 × 47.1	44.0	5	138

Transect count and sweep time assume the 10 ha area split 3 ways, each sub-region taken as square, flown at 8 m/s with a 6 s cost per turn.

5. What the substitution changes, at 40 m

	IMX477	Sizing model	Change
Horizontal field of view	55.30	63.20	-12.5 %
Ground sample distance	1.03	1.21	-14.8 %
Survivor, long axis	164.52	140.11	+17.4 %
Swath width	41.91	49.21	-14.8 %
Line spacing	29.34	34.45	-14.8 %
Transects per drone	7.00	6.00	+16.7 %
Sweep time per drone	195.75	166.93	+17.3 %

A smaller pixel behind the same lens narrows the field and finens the ground sample distance in the same proportion. **Detection improves by 17 % and coverage costs 17 % more sweep time.** Against 250 marks for detection and a 1800 s mission budget, that trade favours the part actually specified — but it was never chosen, and the published figures describe neither camera.

6. Motion blur

Forward motion smears the image by $v\,t_{\mathrm{exp}}$ on the ground. Holding that under one pixel:

$$t_{\mathrm{exp}} \leq \frac{\mathrm{GSD}}{v}$$

AGL (m)	**Max exposure** (ms)	**Shutter**
30	0.97	1/1032
40	1.29	1/774
60	1.94	1/516

`SYS-45` already requires $\leq 1/1000$ s, which clears every altitude with margin. Blur is not a binding constraint; it is Indian daylight driving ISO down rather than exposure up.

7. Tiling

A survivor occupies a few tens of pixels in a 12 MP frame, so the detector is fed overlapping crops rather than the whole image. Tile count depends on pixel *count*, so it is unaffected by the sensor substitution — but the *effective* GSD after downsampling is not.

Downsample	**Frame**	**Tiles**	**Inf./s**	**GSD** (cm/px)	**Survivor** (px)
1×	4056×3040	48	240	1.03	165
2×	2028×1520	12	24	2.07	82

At 2× downsampling and 2 Hz the budget closes at 24 inferences per second, unchanged. The survivor is **82 px** in that downsampled frame — not the 47 px the proposal currently states, which was computed at 60 m on the other sensor.

8. What follows

Nothing here argues for changing the camera. The IMX477 is the better of the two for detection, which is where the marks are. What it argues is that the *documents* should be re-baselined onto the part being bought: the GSD table, the swath and line-spacing figures, the sweep times and the survivor pixel count in the perception section. That work interacts with the unresolved 40 m versus 60 m survey-altitude question, because both move the same quantities, and the two should be settled in one pass rather than two.

Generated file. Regenerate with
`python tools/sizing-model/camera_optics_tex.py` then `pdflatex`.
Every figure above is computed by `tools/sizing-model/camera_optics.py`.